

Clear Research Papers for the Job Recommendation Project

Project: Job Recommendation and Candidate Matching Using Content-Based Filtering and Cosine Similarity

Purpose: These papers are selected to support the Mini Hackathon literature survey and viva. The explanations are simplified so they can be understood and presented easily.

1. Job Recommender Systems: A Systematic Literature Review, Applications, Open Issues, and Challenges

Authors: Duygu Çelik Ertuğrul and Selin Bitirim

Year: 2025

Publication: Journal of Big Data

DOI: 10.1186/s40537-025-01173-y

Full paper: <https://link.springer.com/article/10.1186/s40537-025-01173-y>

What the paper is about

This is a large systematic literature review of Job Recommender Systems. It analyzes 57 research articles and organizes the main approaches used for recommending jobs, including content-based, collaborative, hybrid, knowledge-based and other approaches.

How it connects to your project

Your project uses content-based filtering. This paper gives strong research justification for choosing content-based recommendation as a baseline approach.

Simple viva explanation

We referred to this systematic review to understand the major approaches used in Job Recommender Systems. Our project implements a simpler content-based approach using TF-IDF and cosine similarity.

2. Technical Job Recommendation System Using APIs and Web Crawling

Authors: Naresh Kumar, Manish Gupta, Deepak Sharma, Isaac Ofori

Year: 2022

Publication: Computational Intelligence and Neuroscience, Article 7797548

DOI: 10.1155/2022/7797548

Full paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9239795/>

What the paper is about

This paper describes a technical job recommendation system. It discusses collecting job information, processing job data, and recommending suitable technical jobs. The study uses content-based and collaborative filtering as part of its recommendation approach.

How it connects to your project

It is useful for understanding how a real job recommendation application can be structured. Your implementation is intentionally simpler: candidate profile → TF-IDF → cosine similarity → ranking.

Simple viva explanation

This paper is related to our implementation because it studies automated technical job recommendation. Our project focuses on the content-based part rather than implementing the full hybrid system.

3. Combining Content-Based and Collaborative Filtering for Job Recommendation System: A Cost-Sensitive Statistical Relational Learning Approach

Authors: Shuo Yang, Mohammed Korayem, Khalifeh AlJadda, Trey Grainger, Sriraam Natarajan

Year: 2017

Publication: Knowledge-Based Systems, Volume 136, pages 37–45

DOI: 10.1016/j.knosys.2017.08.017

Full paper: <https://www.sciencedirect.com/science/article/pii/S095070511730374X>

What the paper is about

This paper combines content-based and collaborative filtering using Statistical Relational Learning. It models whether a candidate and a job are a good match and considers the different costs of recommendation errors.

How it connects to your project

It gives you a clear future-scope direction: after establishing a content-based baseline, the system could incorporate user interaction history and other signals through a hybrid recommender.

Simple viva explanation

Our current mini-project is content-based. A future version could combine it with collaborative filtering so previous applications, clicks or preferences could also influence recommendations.

4. Job Recommender Systems: A Review

Authors: Corn   de Ruijt and Sandjai Bhulai

Year: 2021

Publication: arXiv

DOI: 10.48550/arXiv.2111.13576

Full paper: <https://arxiv.org/abs/2111.13576>

What the paper is about

This review studies Job Recommender Systems over the 2011–2021 period and discusses issues such as temporal preferences, reciprocal matching and fairness.

How it connects to your project

It helps explain that real-world job recommendation is more complex than simply finding similar text. These topics can be listed as future improvements to your project.

Simple viva explanation

This review helped us understand broader challenges in job recommendation, while our mini-project focuses specifically on content similarity between candidates and jobs.

Recommended Literature Survey Slide

Paper	Year	Main focus	Relation
Ertu��rul & Bitirim	2025	Systematic review of JRS	Research background
Kumar et al.	2022	Technical job recommendation	Related implementation

Paper	Year	Main focus	Relation
Yang et al.	2017	Hybrid recommendation	Future enhancement
de Ruijt & Bhulai	2021	JRS review and challenges	Broader background

Research gap for the PPT: Existing research includes content-based, collaborative and hybrid job recommendation approaches. This mini-project implements a simple and interpretable content-based baseline using TF-IDF and Cosine Similarity to compare a candidate profile with job postings and rank recommendations.

Important: The papers contain methods beyond your current implementation. In your viva, clearly state that your project currently implements content-based filtering with TF-IDF and cosine similarity; collaborative filtering, semantic embeddings, web crawling and advanced evaluation are possible future extensions rather than current features.